

SATURATED MOBILE B-DIVISORS ON WEAK LOG DEL PEZZO SURFACES

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1. INTRODUCTION

It is the aim of this section to give a rather precise description of mobile b-divisors on a weak log del Pezzo surface:

Theorem 1.1. *Let $(X, B) \rightarrow Z$ be relative weak del Pezzo klt surface pair. Denote by $(X', B') \rightarrow (X, B)$ the terminal model. There are*

- (1) *finitely many normal varieties T_i/Z and Z -morphisms $\varphi_i: X' \rightarrow T_i$,*
- (2) *finitely many normal surfaces Y_j/Z together with projective birational Z -morphisms $h_j: Y_j \rightarrow X$, nonsingular curves Δ_j/Z together with fibrations $\chi_j: Y_j \rightarrow \Delta_j$ defined over Z , and*
- (3) *a bounded algebraic family $\mathcal{F}$ of mobile $\mathbf{A}(X, B)$ -saturated b-divisors on X ,*

such that the following holds: If $\mathbf{M}$ is any mobile $\mathbf{A}(X, B)$ -saturated b-divisor on X , then either

- (a) *$\mathbf{M}$ descends to the terminal model $(X', B') \rightarrow (X, B)$ and the linear system $|H^0(X, \mathbf{M})|$ defines one of the morphisms $\varphi_i: X' \rightarrow T_i$, or*
- (b) *$\mathbf{M}$ descends to one of the Y_j and $|H^0(X, \mathbf{M})|$ defines the morphism $\chi_j: Y_j \rightarrow \Delta_j$, or*
- (c) *$\mathbf{M}$ belongs to the family $\mathcal{F}$.*

(1.1) is used in the proof of the existence of 4-fold flips [1], and in fact (1.1) is an exact restatement of (???) of [1]. In fact the only part of (1.1), which has not already been proved in [2], concerns the extra information contained in parts (2) and (b):

Theorem 1.2. *Let (S, Δ) be a kawamata log terminal pair, where S is a surface and $-(K_S + \Delta)$ is big and nef.*

Then there are finitely many contraction morphisms $f_j: S_j \rightarrow \mathbb{P}^1$, $j = 1 \dots k$, where $\pi: S_j \rightarrow S$ is a birational morphism, with the following property:

Let $\mathbf{M}$ be a mobile and saturated b -divisors of Kodaira dimension one. Then there is an index j such that $\mathbf{M}$ descends to the surface S_j and $\mathbf{M}_{S_j}$ is the pullback of an ample divisor from $\mathbb{P}^1$.

2. MOBILE B-DIVISORS OF KODAIRA DIMENSION ONE

In this section we prove (1.2).

Definition 2.1. Let S be a rational surface and let $\mathbf{M}$ be a mobile b -divisor of Kodaira dimension one. Let C be an irreducible component of the general element of $\mathbf{M}|_S$ and let W be the pencil corresponding to C . We refer to W as the **pencil** associated to $\mathbf{M}$.

Remark 2.2. (1.2) is equivalent to the statement that there are only finitely many pencils associated to any mobile and $\mathbf{A}(S, \Delta)$ -saturated b -divisor on a surface S , such that $K_S + \Delta$ is kawamata log terminal and $-(K_S + \Delta)$ is big and nef.

Lemma 2.3. To prove (1.2), we may assume that the pair (S, Δ) is terminal.

Proof. Obvious, since we are clearly allowed to pass to a terminal model. $\square$

Lemma 2.4. Let (S, Δ) be a terminal pair, where S is a surface and $-(K_S + \Delta)$ is big and nef. Suppose that $\mathbf{M}$ is a mobile, $\mathbf{A}(S, \Delta)$ -saturated b -divisor of Kodaira dimension one. Let $f: T \rightarrow \mathbb{P}^1$ be the graph of the rational map $\phi: S \dashrightarrow \mathbb{P}^1$ determined by W , the pencil associated to $\mathbf{M}$. Let $\pi: T \rightarrow S$ be the natural morphism, and let C be an irreducible component of a general element of W .

Then the following hold

- (1) π is an isomorphism outside a single point $p \in S$.
- (2) There is a unique exceptional divisor E which meets the strict transform C' of C in a single point.
- (3) The log discrepancy of E is at most two.
- (4) C is a smooth elliptic curve.

Proof. Follows from (???) of [2]. $\square$

The next couple of lemmas are dedicated to proving the following result:

Proposition 2.5. Let (S, Δ) be a kawamata log terminal pair, where S is a surface and $-(K_S + \Delta)$ is big and nef.

Then there are finitely complete linear systems V_j , $j = 1 \dots k$, with the following property:

For every mobile and saturated b -divisor $\mathbf{M}$ of Kodaira dimension one, there is an index j such that $V_j = |C|$, where C is a general element of the pencil associated to $\mathbf{M}$.

Lemma 2.6. *Let S be a smooth rational surface, and let C be a smooth elliptic curve, of positive self-intersection.*

If $K_S + C$ is nef, then $C \in |-K_S|$.

Proof. As C is an elliptic curve,

$$\mathcal{O}_C(K_S + C) = \mathcal{O}_C(K_C) = \mathcal{O}_C.$$

On the other hand, as S is rational,

$$H^1(S, \mathcal{O}_S(K_S)) = H^1(S, \mathcal{O}_S) = 0.$$

Considering the long exact sequence associated to the short exact sequence,

$$0 \longrightarrow \mathcal{O}_S(K_S) \longrightarrow \mathcal{O}_S(K_S + C) \longrightarrow \mathcal{O}_C \longrightarrow 0,$$

it follows that $H^0(S, \mathcal{O}_S(K_S + C)) \neq 0$. Thus we may write

$$K_S + C = B,$$

where B is integral and effective. Furthermore $B \cdot C = 0$ as $(K_S + C) \cdot C = 0$. If $B = 0$, then $C \in |-K_S|$ and we are done. Otherwise, by the Hodge Index Theorem, it follows that $B^2 < 0$, which contradicts the fact that $K_S + C$ is nef. $\square$

Lemma 2.7. *Let S be a smooth rational surface, and let C be a smooth elliptic curve, of positive self-intersection.*

Then there is a birational morphism $f: S \longrightarrow S'$, which is an isomorphism in a neighbourhood of C , such that the image C' of C is an element of $|-K_{S'}|$.

Proof. Consider running the $(K_S + C)$ -MMP. We get a birational morphism $f: S \longrightarrow S'$. Note that for every step of this MMP, we contract a -1 -curve Σ . As $(K_S + C) \cdot \Sigma < 0$, $K_S \cdot \Sigma = -1$ and $C \cdot \Sigma$ is an integer, it follows that Σ is disjoint from C , so that f is an isomorphism in a neighbourhood of C .

Replacing the pair (S, C) by the pair (S', C') , where C' is the image of C , we may assume that either

- (i) $K_S + C$ is nef, or
- (ii) we have a Mori fibre space $\pi: S \longrightarrow Z$, such that $-(K_S + C)$ is π -ample.

If (i) holds, then we are done by (2.6). Suppose that (ii) holds. Note that $(K_S + C) \cdot C = 0$, as C is an elliptic curve, so that $K_S + C$ is non-negative on at least one ray of the closed cone of curves. Thus Z must be a curve, and C must be a section of π . But then Z is isomorphic to C , which contradicts the fact that S is rational. $\square$

Lemma 2.8. *Let (X, Δ) be a kawamata log terminal pair, where $-(K_X + \Delta)$ is big and nef.*

Then the closed cone of curves of X is polyhedral. In particular there are only finitely many birational morphisms $f: X \rightarrow Y$.

Proof. By the base point free theorem, there is a birational morphism $f: X \rightarrow Y$, such that if $\Gamma = f_*\Delta$, then $-(K_Y + \Gamma)$ is ample. It follows that there is a small perturbation Δ' of Δ , such that $-(K_X + \Delta')$ is ample, and this result is then an immediate consequence of the cone theorem. $\square$

Proof of (2.5). By (2.8) there are only finitely many birational morphisms $f_i: S \rightarrow S_j$, $j = 1, 2, \dots, k$. For each j , we may write

$$K_S = f^*K_{S_j} + E_j,$$

where E_j is exceptional for f_j . Let $\mathbf{M}$ be a mobile, $\mathbf{A}(S, \Delta)$ -saturated b-divisor, of Kodara dimension one. Let C be an irreducible component of a general element of W , the pencil associated to $\mathbf{M}$. By (2.7), there is an index j , such that $C' = f_j(C) \subset S_j$ is an element of $-K_{S_j}$, where f_j is an isomorphism in a neighbourhood of C . It follows that

$$C = f^*C' = -f^*K_{S_j} = -K_S + E_j.$$

Thus $|C| = |-K_S + E_j|$, so that we may take $V_j = |-K_S + E_j|$, for $1 \leq j \leq k$. $\square$

Lemma 2.9. *Let S be a smooth surface and let $\Delta = aB$, where B is a smooth curve. Let $\pi: T \rightarrow S$ be a birational morphism, which is a sequence of k blow ups with centre p . Let E be the last exceptional divisor extracted by π .*

If the log discrepancy of E is at most two, then either

- (1) $k = 1$, or
- (2) *the centre of each blow up is along the strict transform of B , and*

$$a \geq 1 - \frac{1}{k}.$$

Proof. Suppose that we make t blow ups $f_t: S_t \rightarrow S$ with centre p . I claim that the log discrepancy of the last exceptional divisor E_t

extracted by f_t is at least

$$1 + t(1 - a),$$

with equality iff every blow up is along B .

By induction we may assume the result up to $t - 1$. If the log discrepancy of E_{t-1} is b , then its coefficient is $b - 1$ and so the log discrepancy of E_t is at least

$$2 - a + b \geq 2 - a + (t - 1)(1 - a) = 1 + t(1 - a),$$

with equality iff all blow ups are along B . This completes the induction and the result follows easily. $\square$

Lemma 2.10. *Let S be a smooth surface and let Δ be a divisor, such that $K_S + \Delta$ is kawamata log terminal and $-(K_S + \Delta)$ is big and nef.*

Suppose that $\mathbf{M}$ is a mobile, $\mathbf{A}(S, \Delta)$ -saturated b -divisor of Kodaira dimension one, with associated pencil W . Assume that the base locus of W is not empty. By (2.4) W has a unique base point p . Let C be an irreducible component of a general element of W . Then, either

- (1) $C^2 = 1$.
- (2) p is a singular point of the support of Δ .
- (3) $p \in B$, a component of Δ . Let $V = |C|$ be the complete linear system on S and let U be the restriction of V to B . Then $dp \in U$, where $B \cdot C = d$, whilst the dimension of U is at most $d - 1$.

Proof. By (2.4) the log discrepancy of the component E of $\mathbf{A}_T(S, \Delta)$ which meets C' is at most two. Suppose that $k = C^2 > 1$. Again by (2.4) it follows that $p \in \Delta$. Suppose that p is not a singular point of the support of Δ . Let B be the unique component of Δ which contains p . By (2.9) π blows up along B , k times. It follows that

$$k = C^2 \leq C \cdot B.$$

On the other hand,

$$\begin{aligned} 0 &\leq C \cdot -(K_S + \Delta) \\ &\leq C \cdot -(K_S + aB) \\ &= k - aB \cdot C \\ &\leq k - \frac{(k - 1)}{k} B \cdot C, \end{aligned}$$

where we used (2.9) to get from the penultimate line to the last line. Thus

$$(k - 1)B \cdot C \leq k^2.$$

It follows that $B \cdot C \leq k + 1$. Thus $B \cdot C = k$ or $k + 1$. Let B' be the strict transform of B on T .

Suppose that $B \cdot C = k$. Then $B' \cdot C' = 0$, and so B' is contained in a fibre of g . It follows that B' is a fibre of g . It follows that $B \cap C$ is the single point p , so that $B \cap C = kp$. Thus $kp \in U$. In this case V has dimension d , but the linear map

$$V \longrightarrow U$$

has a one dimensional kernel, so that the dimension of U is at most $d - 1$.

Now suppose that $B \cdot C = k + 1$. Then $B' \cdot C' = 1$, so that B' is section of g . But then $B' \simeq \mathbb{P}^1$, and so $B \simeq \mathbb{P}^1$, as p is a smooth point of B . It follows that $C \cap B$ consists of at most two points, p and q , and $B \cap C = (d - 1)p + q$. Thus $V \subset |q| + (d - 1)p$. Now the restriction map

$$W \longrightarrow V,$$

must have one dimensional image, else there would be an element $C \in W$ which contains B , a contradiction. Thus q cannot be a base point of V and so $dp \in V$. In this case the dimension of U is at most $k = d - 1$. $\square$

Lemma 2.11. *Let B be an integral curve. Let V be a linear system on B of degree d , which contains dp , for infinitely many points $p \in B$.*

Then $B \simeq \mathbb{P}^1$ and V is the full linear system of divisors of degree d .

Proof. Suppose first that the arithmetic genus of B is greater than zero. Then the morphism

$$B \longrightarrow \text{Pic}(B) \quad \text{which sends} \quad p \longrightarrow \mathcal{O}_B(p),$$

is finite. Since the morphism

$$\text{Pic}(B) \longrightarrow \text{Pic}(B) \quad \text{which sends} \quad L \longrightarrow L^{\otimes d},$$

is also finite, it follows that the morphism

$$B \longrightarrow \text{Pic}(B) \quad \text{which sends} \quad p \longrightarrow \mathcal{O}_B(dp),$$

is finite.

Thus B has arithmetic genus zero, so that B is isomorphic to $\mathbb{P}^1$. Now under the identification of $\mathbb{P}^d \simeq \mathbb{P}(H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1})^*)$, the set of divisors of the form dp is precisely a rational normal curve of degree d . Since the rational normal curve of degree d spans $\mathbb{P}^d$, the result is clear. $\square$

Proof of (1.2). Suppose not. Then we could find infinitely many mobile and $\mathbf{A}(S, \Delta)$ -saturated b-divisors $\mathbf{M}_j$ of Kodaira dimension one, such that the associated pencils, $W_1, W_2, \dots$ are all distinct. We will derive a contradiction.

By (2.3) we may assume that S is smooth. Let C_j be a general element of W_j . Possibly passing to a subsequence, we may assume that the linear system $V = |C_j|$ is fixed, by (2.5). By (2.6), $0 \leq C_j^2 \leq 9$, so that we may also assume that $k = C_j^2$ is fixed.

The proof proceeds by induction on k . If $k = 0$ (strictly speaking $C^2 > 0$, but we consider this possibility for the purposes of induction), then $W_j = V$ and there is nothing to prove. Otherwise suppose that $k > 0$. Let p_j be the unique base point of the linear system W_j . If for infinitely many j , $p = p_j$ is fixed, then we could replace S by its blow up at p , and proceed by induction on k (at this point k might be zero).

By (2.10) it follows that that we may assume that p_j belongs to the smooth locus of the support of Δ . As Δ has only finitely many components, it follows that we may assume that p_j belongs to a fixed component B of Δ . It follows by (2.10) and (2.11) that B is a copy of $\mathbb{P}^1$ and V cuts out the complete linear series on B , which is impossible by (2.10). $\square$

3. PROOF OF (1.1)

Proof of (1.1). (???) of [2] treats the case when the Kodaira dimension of $\mathbf{M}$ is not one. (1.2) treats the case when the Kodaira dimension is one and Z is a point. By (???) of [2] if C is an irreducible component of the general element of $|\mathbf{M}|_S$, then we may assume that C is complete. As C is mobile, the only remaining case is when Z is a curve, and C is a component of the general fibre of $X \rightarrow Z$. But then the base locus of $|\mathbf{M}|_S$ is empty and there is nothing to prove. $\square$

REFERENCES

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